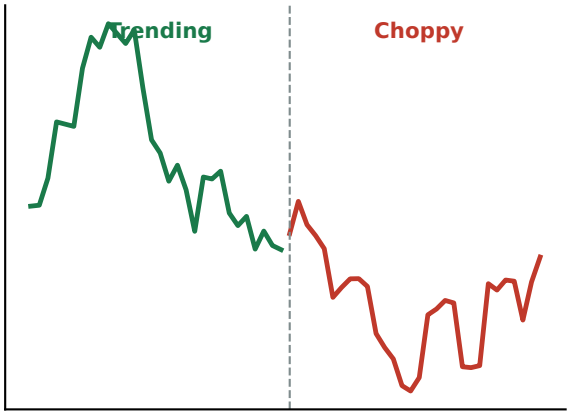
# Regime Ensemble

Detecting market conditions to reduce risk — not just maximise returns

SPY · 2000–2025 · v3.1

## What is a market regime?




*Markets switch between trending and choppy phases.*
*Spot the switch early and you can avoid the worst of it.*

## Why does it matter?

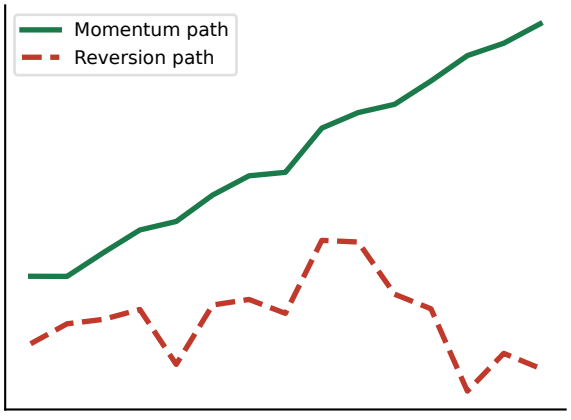
Stay invested when momentum is real,

step aside when conditions turn choppy.

The goal is not to beat the market every year —

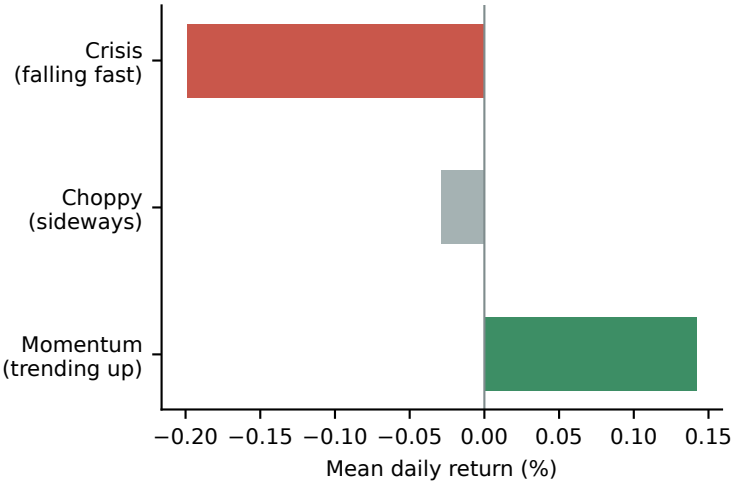
it's to avoid catastrophic drawdowns.

## Detector 1 — Geometric (path shape)




*Measures how straight the last 15 days of price movement was.*
*A straight path → trending. A zigzag → mean-reverting.*

## Detector 2 — Markov (hidden state)




*A statistical model that learns 3 hidden market states from return patterns.*
*Identifies crisis conditions and suppresses buy signals automatically.*

## Headline Results (SPY 2000-2025, zero transaction costs)

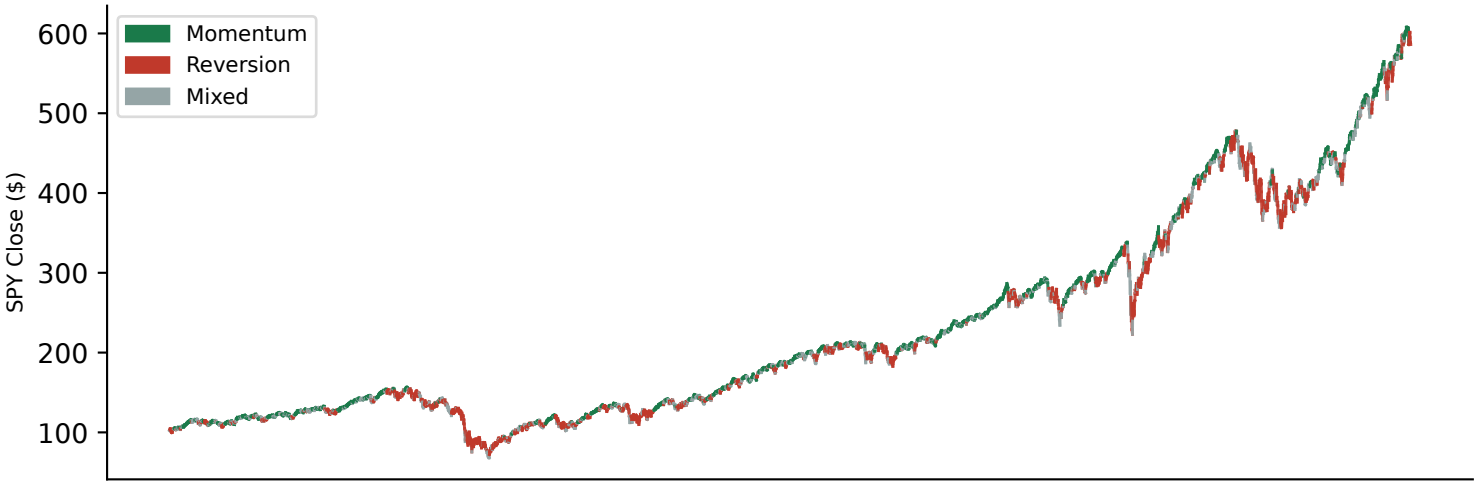
| | CAGR | Sharpe Ratio | Max Drawdown | T-stat (momentum) |
|---|---|---|---|---|
| Strategy (Regime Ensemble) | **+5.7%** | **0.68** | **-16.7%** | **3.13** |
| Buy & Hold (benchmark) | +8.6% | 0.44 | -56.5% | 2.02 |

⚠ At 10bps round-trip costs, Sharpe drops to 0.28. Strategy breaks even around 20bps. Transaction costs are material — see page 3.

# Models in Action — SPY 2000-2025 (Real Data)

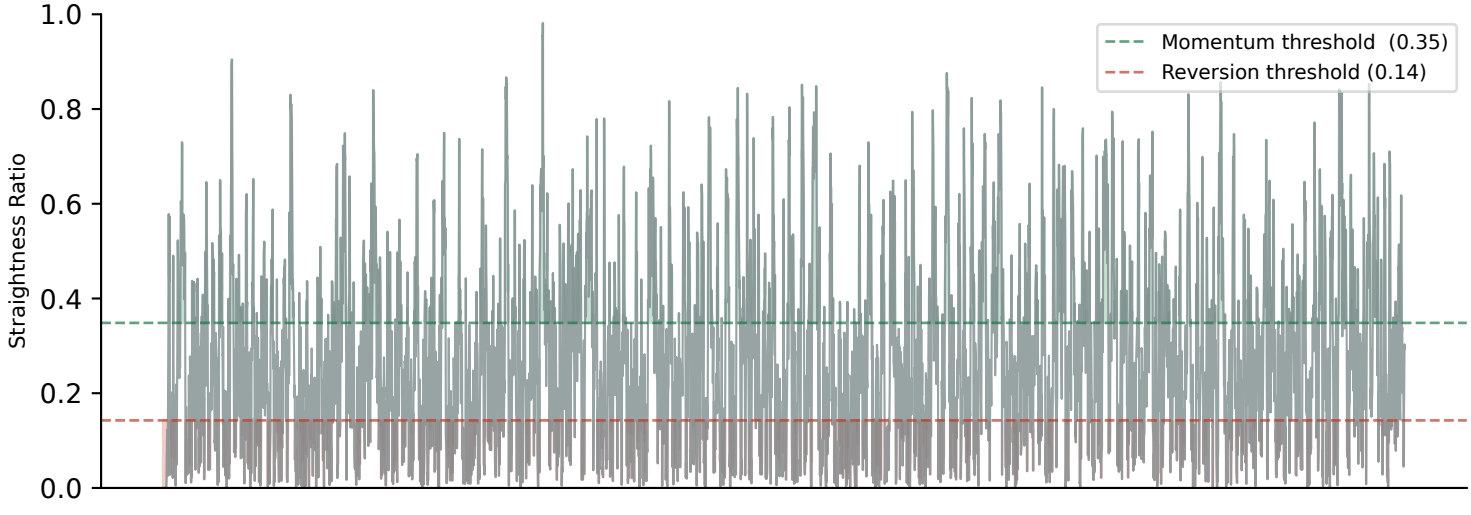
Three panels: price coloured by regime, geometric detector signal, Markov probabilities

## SPY Price — Coloured by Ensemble Regime



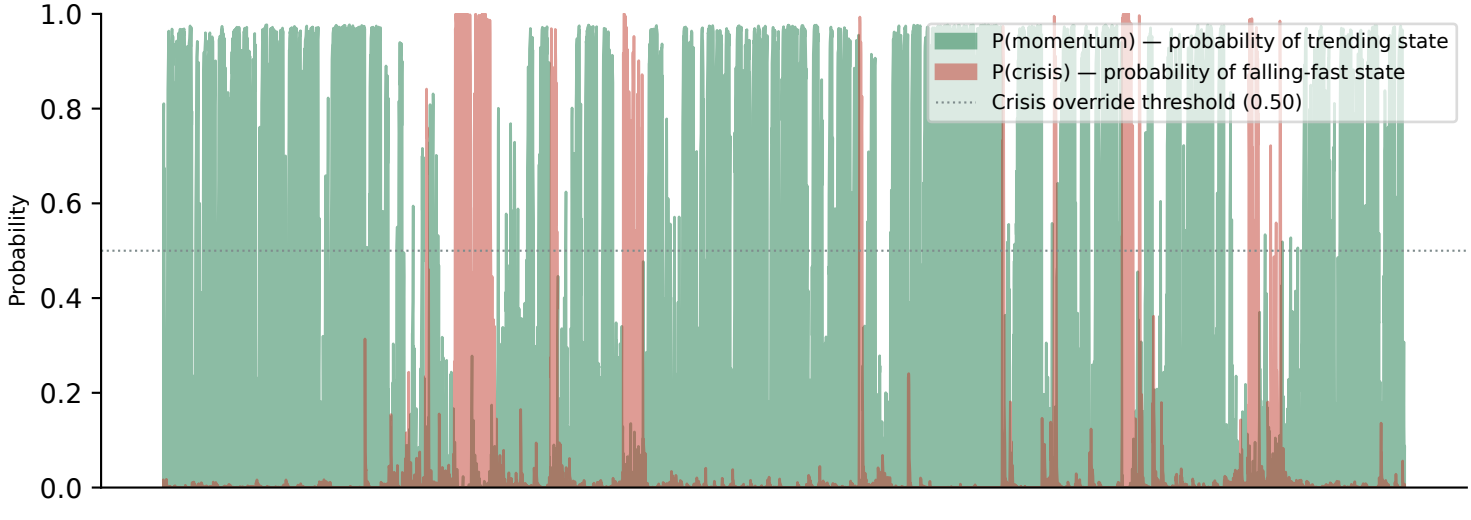

*Green = momentum (hold). Red = reversion (cash). Grey = mixed (cash).*
*Green = momentum, red = crisis/reversion, grey = mixed. Covers dot-com, GFC 2008, COVID 2020, 2022 bear market.*

## Geometric Detector — Straightness Ratio (15-day window)




*Near 1.0 = price moved in a straight line (momentum). Near 0.0 = price zigzagged (mean-reversion).*
*Thresholds are adaptive percentiles — no fixed values to hand-tune.*

## Markov Detector — Filtered State Probabilities (k=3 regimes)

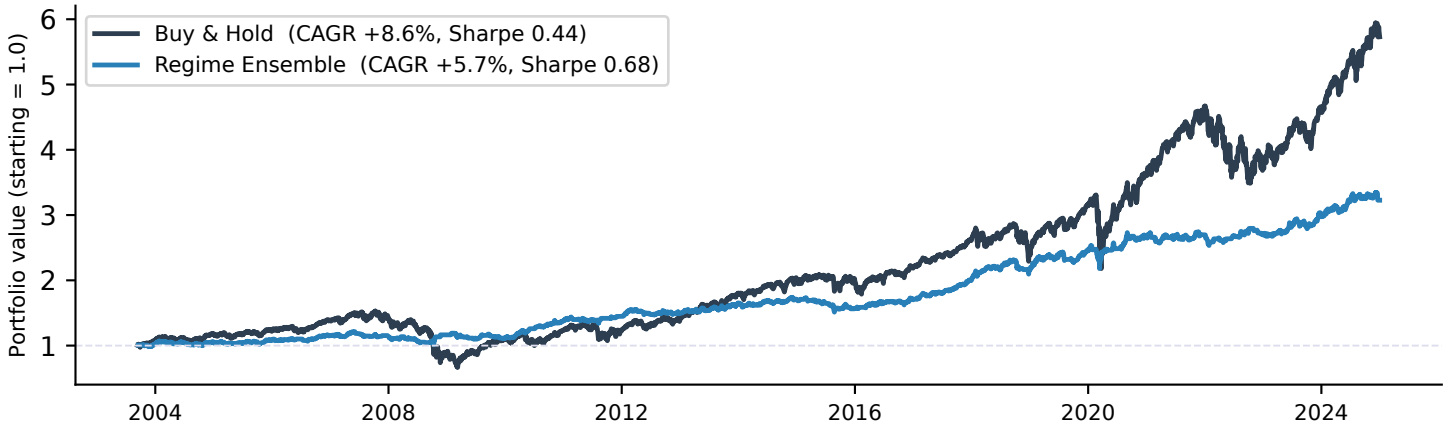



*When P(crisis) crosses 0.50, buy signals are suppressed regardless of other indicators.*
*Filtered probabilities only — no look-ahead bias.*

# Results & Limitations

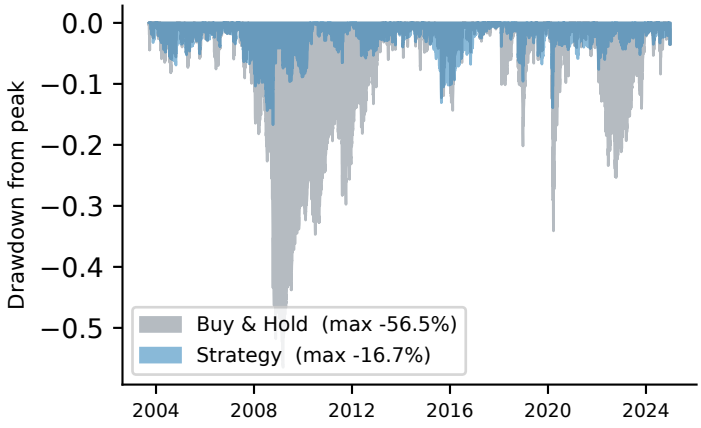
Equity curves, drawdowns, transaction cost sensitivity, and what to watch out for

## Equity Curves — Strategy vs Buy & Hold (0 bps costs)



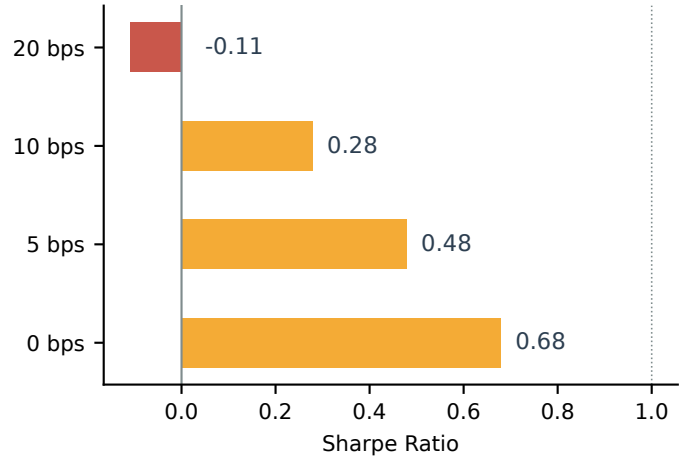

*Strategy Sharpe (0.68) beats buy-and-hold (0.44) with far lower drawdown (-16.7% vs -56.5%).*
*CAGR trails B&H (+5.7% vs +8.6%) -- the edge is risk-adjusted return, not raw return.*

## Drawdown Comparison




*Strategy reduces drawdowns across all major crashes (dot-com, GFC, COVID, 2022).*

## Transaction Cost Sensitivity




*~22 regime switches/year means costs add up fast.*
*At 20bps round-trip, strategy is unprofitable.*

## Key Limitations — Read Before Drawing Conclusions

⚠ **Longer sample, but single asset**
2000-2025 covers dot-com, GFC, COVID -- but it is still only SPY. Cross-validate on other tickers.

⚠ **Transaction costs are material**
At 10bps round-trip, Sharpe drops to 0.28. Breaks even around 20bps. Manage turnover.

⚠ **In-sample threshold calibration**
Regime thresholds were tuned on the full dataset. Real-time use requires expanding-window recalibration.

⚠ **Reversion signal is weak**
Overall strategy T=3.13 (p=0.002). Reversion p=0.73 -- not significant. Don't short on it.

⚠ **Single asset only**
Everything here is SPY. Regime structure may not generalise to other assets, sectors, or markets.